Dynamic Source-Aware Transfer Learning for Vehicular Platoon Predictive Control

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***Abstract*—The allocation of computational resources in the vehicular platoon is susceptible to communication topology, particularly under complex scenarios, such as off-ramps. This article proposes a dynamic source-aware transferable data-driven control (DSTDC) strategy to optimize computing resource allocation under the off-ramps scenario. A data-driven predictor is established to characterize the powertrain structure of vehicular platoon. To improve the computational efficiency of platoon control algorithm, a transfer learning method is constructed. Specifically, the knowledge of vehicle in the source domain is transferred to vehicles in the target domain, and the latter just should update the knowledge in terms of the individual dynamic characteristics and tasks. Furthermore, an adaptive domain switching mechanism is developed to address the issue of source domain vehicle departure in the off-ramps scenario. Here, the source domain vehicle will be subject to reevaluation in terms of the computing performance, data accumulation and driving stability. Finally, a platoon experiment platform is conducted, and the results demonstrate the effectiveness of the proposed DSTDC strategy.**

***Index Terms*—Data-driven model, off-ramps, predictive control, transfer learning, vehicular platoon.**

I. Introduction

THE evolution of connected and autonomous vehicles (CAVs) marks a transformative shift in intelligent transportation system, improving road safety and efficiency [1]. The exploration of vehicle platoons could be traced back to the program on advanced technology for the highway project conducted in California and USA [2]. The vehicular platoon involves the synchronization of CAVs to travel on the complex traffic scenarios, leveraging advanced communication technologies and autonomous driving algorithms. It requires the platoon to maintain a safe intervehicle spacing from the surrounding vehicles [3]. Therefore, a suitable control strategy could ensure the efficient operation of CAVs, thereby the exploitation of the platoon control approaches has triggered a widespread research upsurge [4]. Despite these advancements, the burgeoning data generation and processing requirements impose significant computational burdens [5]. As the scale and complexity of vehicular networks expand, the imperative for robust algorithms and efficiently onboard hardware [6].

The investigation of control algorithms is paramount to guarantee the safety and efficiency of vehicular platoon [7], [8]. Among the abundance of platoon control methodologies, the data-driven control method offers distinct advantages over traditional model-based control approaches [9]. It should be noted that traditional model-based control methods rely on static models and predefined rules, often falling short in dynamic environments [10]. In contrast, data-driven approaches utilize real-time and historical data to inform decision-making processes [11]. This allows for a more flexible and adaptive control system capable of responding to the complex and dynamic nature of actual driving conditions. As a representative data-driven method, an artificial neural network has been widely used in data prediction and control of vehicles [12], [13]. However, this approach essentially constitutes a closed box model, the mechanism of vehicle is difficult to formulate clearly. Afterward, the Koopman operator theory [14] gave a data-driven way to represent a nonlinear dynamical system by a linear expression in an infinite dimensional space. This method has high computational complexity when dealing with high-dimensional complex systems, which may lead to bottlenecks in real-time performance and computing resources. Similarly, the prominent reinforcement learning control method is also subject to the aforementioned computational resource limitations [15], [16]. These problems must be adequately addressed before the large-scale deployment of CAVs, thus, they have drawn additional research attention.

As the scale of the platoon expands, computational costs also increase correspondingly. It would generate the need to manage

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and coordinate a larger number of vehicles, which requires more complex calculations and data handling to maintain system performance and efficiency [17], [18]. The implementation of the aforementioned data-driven schemes in vehicular chips is challenging due to their significant computational demands. To reduce computational costs, various model compression techniques such as weight pruning, parameter quantization and model distillation triggered widespread attention [19], [20]. These techniques aim to maintain the performance of data-driven models within an acceptable accuracy range while substantially reducing parameter counts and computational complexity [21]. Nonetheless, these methods may require additional resources to execute the compression algorithms, potentially increasing the overall computational burden. Fortunately, the transfer learning method has emerged as a feasible solution for mitigating computational burdens in industrial applications [22], [23]. This technique leverages knowledge from the source domain to improve convergence performance in similar target domains. Recent research has explored selective layer freezing and task-specific subnetwork extraction to minimize unnecessary computations while maintaining transfer performance [24]. However, this method has not yet received sufficient attention in vehicle platoons. Therefore, how to develop an efficient control strategy with transfer learning for vehicular platoons is an ongoing research topic.

For several specific traffic scenarios, e.g., off-ramps and intersections, several CAVs would leave the platoon, leading to the platoon topology being split [25], [26]. The rationale behind this phenomenon is communication topology switching, it necessitates CAVs to adjust the control strategy to ensure driving safety and traffic efficiency. To address this issue, Zheng et al. [27] developed a distributed linear feedback control law to investigate the influence of communication topology on the internal stability and scalability of the vehicle platoon. Similarly, an event-triggered control method was put forward to solve the vehicle platoon tracking problem under random topologies communication [28]. Furthermore, a novel dynamic event-triggered mechanism was introduced, and it published the sampled data packets of each vehicle to the intervehicle network [29]. However, these schemes aim to mitigate communication disruptions caused by delays or network attacks. Different from the above communication faults, communication interruptions refer to the temporary or permanent failure of communication links, resulting in a complete halt in information transmission [30]. In other words, CAVs leaving the platoon would trigger the communication interruptions between source vehicles and target vehicles. Therefore, how to provide an available strategy to mitigate the adverse effects of topology switch ON the off-ramp road.

The research mentioned above in modeling focuses on complex neural network-based data-driven models. It poses a significant computational burden for vehicular onboard hardware. Furthermore, the existing transfer learning method cannot mitigate the adverse effects of topology switch ON the off-ramps road. All these factors motivate us to construct an efficient control scheme for vehicular platoon. The main contributions of this article are summarized below.

1) A subspace-based data-driven platoon model is developed to alleviate the adverse effects of uncertain dynamics.
2) A transfer learning method is implemented to reduce the computational burden of platoon.
3) A dynamic source-aware mechanism is proposed to mitigate the adverse effects of losing the source vehicle on the off-ramp road.

This article is organized as follows. Section II illustrates the vehicular platoon model. Section III designs the control strategy. Field experiments are carried out in Section IV. Experimental results are analyzed in Section V. Section VI draws the conclusion.

Notations: $\|\cdot\|_{\mathrm{II}}$ stands for the Frobenius norm. $\triangle$ is the error between the current vector and the reference vector. $|A|$ denotes the absolute of A. $\|\cdot\|^2$ represents the L_2 norm. A^+ and A^- are the positive and negative ideal solution. The average value of matrix A is denoted as $\bar{A}$.

II. Preliminaries

A. Vehicular Platoon Modeling

The vehicular platoon has M CAVs with heterogeneous dynamic structures. To formulate a concise model, several reasonable hypotheses are given: 1) the vehicle is traveling on a dry and flat road, with negligible longitudinal slip of the tires; and 2) the vehicle body is symmetric and rigid [27].

The nonlinear longitudinal dynamics of vehicle i $(i \in M)$ are simplified as follows:

$$\begin{cases} \dot{s}_i(t) = v_i(t) \\ \dot{v}_i(t) = \frac{1}{m_i}(\eta_i \frac{T_i(t)}{R_i} - C_{A,i} v_i^2(t) - m_i g f_i) \\ \tau_i \dot{T}_i(t) + T_i(t) = T_{i,\mathrm{des}}(t) \end{cases} \tag{1}$$

where $s_i(t)$ and $v_i(t)$ denote the position and velocity of vehicle i at the continuous time t. m_i is the vehicle mass. $C_{A,i}$ represents the lumped aerodynamic drag coefficient. g_i stands for the acceleration due to gravity. f_i is the coefficient of rolling resistance, $T_i(t)$ denotes the actual driving/braking torque. $T_{i,\mathrm{des}}(t)$ is the desired driving/braking torque. τ_i stands for the inertial delay of vehicle longitudinal dynamics. R_i represents the tire radius.

The feedback linearization technique is introduced to convert the nonlinear model into a linear one. Then, we obtain a linear model for vehicle longitudinal dynamics

$$\tau_i \dot{a}_i(t) + a_i(t) = u_i(t) \tag{2}$$

where a_i denotes the acceleration. u_i is the control input after linearization.

For the vehicular platoon, each vehicle should maintain a desired intervehicular spacing h^* and velocity v^*, where $h_i^*(k) = h_i(k) - \triangle h_i(k)$, $h_i(k) = s_{i-1}(k) - s_i(k)$ and $v_i^*(k) = v_i(k) - \triangle v_i(k)$. In the desired equilibrium state (h^*, v^*), the platoon error model is defined as follows:

$$\begin{cases} \triangle \dot{h}_i(k) = \triangle v_{i-1}(k) - \triangle v_i(k) \\ \triangle \dot{v}_i(k) = a_{i-1}(k) - a_i(k). \end{cases} \tag{3}$$

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According to (1)–(3), the platoon error system is derived as follows:

$$\begin{aligned} x_i(k+1) &= A_i x_i(k) + B_i u_i(k) + D_i a_{i-1}(k) \\ y_i(k+1) &= C_i x_i(k). \end{aligned} \tag{4}$$

$A_i = \begin{bmatrix} 1 & -T_s & -\tau_i \\ 0 & 1 & T_s \\ 0 & 0 & (\tau_i - T_s)\tau_i \end{bmatrix}$, $B_i = \begin{bmatrix} 0 \\ 0 \\ T_s/\tau_i \end{bmatrix}$, $D_i = \begin{bmatrix} 0 \\ -T_s \\ 0 \end{bmatrix}$, where $x_i(k) = [\triangle h_i(k), \triangle v_i(k), a_i(k)]^T$. C_i denotes an identity matrix and y_i is the output of the system states. T_s stands for the sampling period. a_{i-1} is regarded as a disturbance. Such this model has been commonly employed in vehicle platoon studies [31], [32].

B. Graph Theory

In the vehicle platoon, the data of platoon is transmitted through the vehicular ad-hoc network. The interaction topology of the platoon is represented by a weighted directed graph $G = \{\ell, \varpi, \mathcal{W}\}$, where $\ell = \{\upsilon_1, \upsilon_2, \ldots, \upsilon_N\}$ denotes the set of vehicles, $\varpi \subseteq \{(\upsilon_i, \upsilon_j) \mid \upsilon_i, \upsilon_j \in \ell, i \neq j\}$ represents the edge set, $\mathcal{W} = [w_{ij}] \in \mathcal{R}^{N\times N}$ corresponds to the weighted adjacency matrix [33].

The edge $e_{ij} = (\upsilon_i, \upsilon_j)$ in the graph G represents the communication link between vehicle υ_i and its neighbor υ_j. The set of neighbors for vehicle υ_i is expressed by $N_i = \{\upsilon_j \in \ell, e_{ij} \in \varpi\}$. The communication channel between the ith vehicle to the jth vehicle is represented by edge e_{ij}. Thus, the corresponding weight is defined as follows:

$$w_{ij} = \begin{cases} 0, & i = j \text{ or } e_{ij} \notin \varpi \\ a_{ij} > 0, & j \notin E \text{ and } e_{ij} \in \varpi \\ b_j > 0, & j \in E \text{ and } e_{ij} \in \varpi \end{cases}$$

where a_{ij} denotes the interaction weight among the target domain vehicles and source domain vehicles, b_j stands for the interaction weight among the target domain vehicles. The degree matrix of the graph G is defined as $\mathcal{F} = \mathrm{diag}\{f_1, f_2, \ldots, f_n\}$, the diagonal element is $f_i = \sum_{j=1}^{N} w_{ij}$. Additionally, the Laplacian matrix $\mathcal{L}$ of directed graph G is expressed as $\mathcal{L} = \mathcal{F} - \mathcal{W}$.

The vehicular platoon may suffer from the source domain vehicle, and the dynamic source-aware mechanism could provide a reliable solution [34]. A group of directed graph $G_{\sigma(t)} \in \{G_1, G_2, \ldots, G_g\}$ is used to characterize different topologies under communication failures.

III. DSTDC Strategy Design

Fig. 1 presents the framework of the proposed dynamic source-aware transferable data-driven control (DSTDC) strategy. Specifically, the date-driven model is employed to construct the subspace predictor, which could generate the optimization predictive sequences. Based on these predictive sequences, a dynamic source-aware mechanism is designed to assign a fresh source vehicle under off-ramps scenarios. Finally, a transfer learning approach is developed, and the date-driven model and controller hyperparameter of vehicles in the source domain are transferred to vehicles in the target domain, thereby reducing the calculation burden.

A. Transfer Learning

For the distributed platoon control framework, each CAV should build an individual data-driven model using a large historical dataset. This task imposes a significant computational burden, particularly for large-scale vehicles within traffic systems. Transfer learning, leveraging feature sharing and model parameter transfer, plays a crucial role in improving computational efficiency [35]. This approach entails applying knowledge acquired in the source domain ($\mathcal{D}_S$) to the target domain ($\mathcal{D}_T$), thereby improving task performance in the target domain.

Here, the leader-predecessor following (LPF) communication topology is implemented in the platoon. The source domain vehicles and target domain vehicles exhibit similar dynamic characteristics and perform analogous tasks, thus ensuring minimal disparity during the transfer process. However, we acknowledge the inherent differences between vehicles and tasks. Therefore, such discrepancies could be addressed using domain adaptation techniques to eliminate these variations, thereby ensuring the effectiveness of the transfer process [36].

The domain $\mathcal{D}_S$ of platoon involves feature space $\mathcal{S}_s$ and the marginal probability distribution function $\mathcal{P}(s)$ is described as follows:

$$\mathcal{D}_S = \{\mathcal{S}_s, \mathcal{P}(s)\}, s \in \mathcal{S}_s. \tag{5}$$

Accordingly, the domain $\mathcal{D}_T$ comprises the label space $\mathcal{S}_r$ and the predictive function $\mathcal{F}(s)$. That is

$$\mathcal{D}_T = \{\mathcal{S}_r, \mathcal{F}(s)\}, s \in \mathcal{S}_r. \tag{6}$$

Given the differences in feature distributions between $\mathcal{D}_S$ and $\mathcal{D}_T$, it becomes essential to conduct classification and prediction on the unlabeled datasets within the target domain $\mathcal{D}_S$. Therefore, a Kullback-Leibler divergence method is utilized to measure the similarity between $\mathcal{D}_S$ and $\mathcal{D}_T$ [37]. This can be described by the following equation:

$$\mathcal{D}_{KL}(P \parallel Q) = \sum_{s\in\mathcal{S}_r} P(s) \log \frac{P(s)}{Q(s)} \tag{7}$$

where $P(s)$ represents the probability distribution of the original unknown data. $Q(s)$ stands for the generated approximate probability distribution.

With this in mind, the principles of transfer learning applied in vehicle platoon are detailed in Fig. 1. To be more specific, the knowledge from CAV_1, comprising pretrained data-driven models and hyperparameters, serves as the source domain. Then, the knowledge is transferred to the other vehicles (CAV_i, $i \geq 2$), referred to as the target domain. It should be noted that target domain vehicles only need to update the knowledge according to the individual dynamic characteristics and tasks. In this way, the initial optimization process of data-driven modeling could be omitted, thereby enhancing the convergence performance of platoon.

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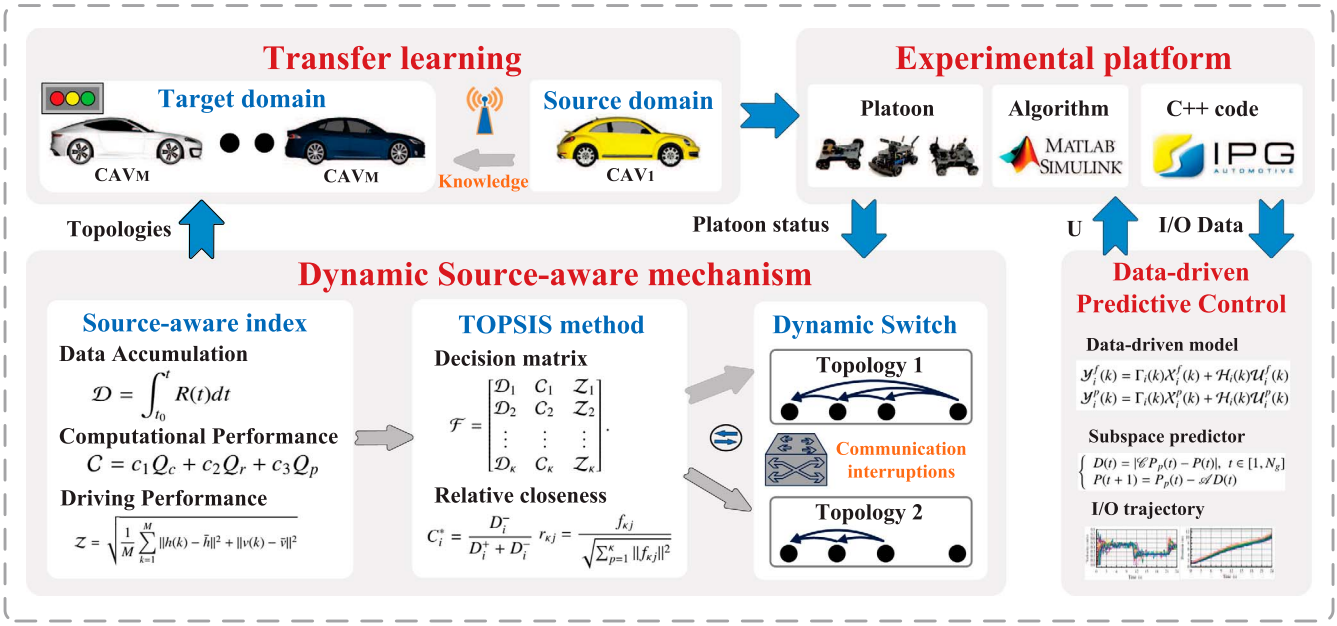

Fig. 1. Framework of DSTDC for platoon.

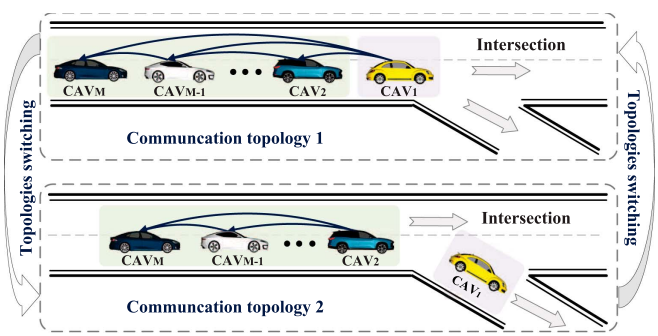

Fig. 2. Dynamic source-aware mechanism of platoon.

B. Dynamic Source-Aware Mechanism

As shown in Fig. 2, the source domain vehicle (CAV_1) would leave the platoon under off-ramp road, resulting in the target domain vehicles losing the guidance of source domain vehicle. In this part, a technique for order of preference by similarity to ideal solution (TOPSIS) approach is utilized to select the novel available source domain vehicle in terms of the data accumulation, computing performance and driving performance of each vehicle.

1) Data Accumulation: The amount of historical driving data determines the selection of source domain vehicle. Here, the driving data is collected from the moment that the vehicle joins the task. Mathematically, this can be expressed as follows:

$$\mathcal{D} = \int_{t_0}^{t} R(t)dt \tag{8}$$

where $\mathcal{D}$ denotes the total driving data, $R(t)$ is the rate of data generation at time t. t_0 stands for the time when the vehicle joined the platoon task.

2) Computational Performance: The computational performance of onboard computers allows for the rapid processing of large datasets, thereby enabling the development of more accurate and reliable models

$$\mathcal{C} = c_1 Q_c + c_2 Q_r + c_3 Q_p \tag{9}$$

where $\mathcal{C}$ denotes the computational performance score. Q_c is the index of processor speed. Q_r stands for the index of memory capacity. Q_p represents the index of data processing rate. c_1, c_2, and c_3 are weight coefficients.

3) Driving Performance: The root mean square error (RMSE) of velocity and spacing is used to evaluate the driving performance. Stable spacing ensures efficient traffic flow and reduces collision risks, while velocity regulation is crucial for maintaining the stability of the platoon. Then, we have

$$\mathcal{Z} = \sqrt{\frac{1}{M}\sum_{k=1}^{M} \|h(k) - \bar{h}\|^2 + \|v(k) - \bar{v}\|^2} \tag{10}$$

where $\mathcal{Z}$ is the driving performance score.

To comprehensively analyze the above indicators, the TOPSIS method provides a feasible solution, see Behzadian et al. [38] for details. The rationale behind this scheme is to calculate the relative distance between the highest risk indicators and current risk indicators.

Based on (8)–(10), a decision matrix $\mathcal{F}$ with κ topology and three evaluation indicators could be derived as follows:

$$\mathcal{F} = \begin{bmatrix} \mathcal{D}_1 & \mathcal{C}_1 & \mathcal{Z}_1 \\ \mathcal{D}_2 & \mathcal{C}_2 & \mathcal{Z}_2 \\ \vdots & \vdots & \vdots \\ \mathcal{D}_\kappa & \mathcal{C}_\kappa & \mathcal{Z}_\kappa \end{bmatrix}. \tag{11}$$

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By normalizing the decision matrix $\mathcal{F}$, such that

$$r_{\kappa j} = \frac{f_{\kappa j}}{\sqrt{\sum_{p=1}^{\kappa} \| f_{\kappa j} \|^2}} \tag{12}$$

where $f_{\kappa j} \in \mathcal{F}$, $p \in \{1, \ldots, \kappa\}$ and $j \in \{1, 2, 3\}$.

Assign different weights to the criteria based on their importance, forming the weighted normalized decision matrix $v_{\kappa j}$. According to the properties of the positive ideal solution and the negative ideal solution, we can obtain the Euclidean distance of each alternative D_i. Then, the relative closeness of each alternative C_i^* is deduced as follows:

$$C_i^* = \frac{D_i^-}{D_i^+ + D_i^-} \tag{13}$$

where $D_i^+ = \sqrt{\sum_{j=1}^{3} (v_{\kappa j} - v_j^+)^2}$ and $D_i^- = \sqrt{\sum_{j=1}^{3} (v_{\kappa j} - v_j^-)^2}$.

Until now, we can determine the value of C_i^*, which serves to assess the feasibility of various topologies. A higher C_i^* signifies a greater alternative. With this in mind, the vehicle with a higher C_i^* is deemed most appropriate for serving as the source domain vehicle.

C. Data-Driven Predictive Control

Data-driven predictive control approach offers several distinct advantages as an advanced control strategy in various industrial applications. The subspace-based data-driven predictive control method uniquely integrates the structural advantages of state-space modeling with the flexibility of data-driven approaches. By leveraging subspace identification techniques, it effectively extracts system dynamics from limited input–output data without relying on explicit parametric models, thereby mitigating issues related to insufficient data samples [39]. Given the significant parameter uncertainty related to complex vehicle structures, this method is utilized to construct the data-driven model.

Here, the data-driven model can be formulated using Hankel matrices by stacking the collected input/output (I/O) trajectories from experiments. That is

$$\mathcal{Y}_i^f(k) = \Gamma_i(k)\mathcal{X}_i^f(k) + \mathcal{H}_i(k)\mathcal{U}_i^f(k) + \mathcal{M}_i(k)\mathcal{W}_i^f(k) \tag{14a}$$

$$\mathcal{Y}_i^p(k) = \Gamma_i(k)\mathcal{X}_i^p(k) + \mathcal{H}_i(k)\mathcal{U}_i^p(k) + \mathcal{M}_i(k)\mathcal{W}_i^p(k) \tag{14b}$$

with

$$\mathcal{Y}_i^f(k) = \begin{bmatrix} y_i(k+L) & \cdots & y_i(k+N) \\ \vdots & \ddots & \vdots \\ y_i(k+2L-1) & \cdots & y_i(k+N+L-1) \end{bmatrix}$$

$$\mathcal{Y}_i^p(k) = \begin{bmatrix} y_i(k) & \cdots & y_i(k+N-L) \\ \vdots & \ddots & \vdots \\ y_i(k+L-1) & \cdots & y_i(k+N-1) \end{bmatrix}$$

$$\mathcal{X}_i^p(k) = \begin{bmatrix} x_i(k) \\ \vdots \\ x_i(k+N-L) \end{bmatrix}^T, \mathcal{X}_i^f(k) = \begin{bmatrix} x_i(k+L) \\ \vdots \\ x_i(k+N) \end{bmatrix}^T$$

$$\Gamma_i(k) = \begin{bmatrix} C_i(k) & C_i(k)A_i(k) & \cdots & C_i(k)A_i^{L-1}(k) \end{bmatrix}^T$$

$$\mathcal{H}_i(k) = \begin{bmatrix} D_i(k) & C_i(k)B_i(k) & \cdots & C_i(k)A_i^{L-2}(k)B_i(k) \\ 0 & D_i(k) & \cdots & C(k)A_i^{L-3}(k)B_i(k) \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & D_i(k) \end{bmatrix}^T$$

$$\mathcal{M}_i(k) = \begin{bmatrix} I & C_i(k)\mathcal{E}_i(k) & \cdots & C_i(k)A_i^{L-2}(k)\mathcal{E}_i(k) \\ 0 & I & \cdots & C_i(k)A_i^{L-3}(k)\mathcal{E}_i(k) \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & I \end{bmatrix}^T$$

$\mathcal{U}_i^f(k), \mathcal{U}_i^p(k), \mathcal{W}_i^f(k)$ and $\mathcal{W}_i^p(k)$ have similar forms to $\mathcal{Y}_i^f(k)$, $\mathcal{Y}_i^p(k)$ and $\mathcal{M}_i(k)$. The subscripts p and f represent the past and future states of corresponding matrices. N and L denote the prediction horizon and excitive order.

Subsequently, the data-driven model (14) is utilized to develop the linear subspace predictor equations. By solving the following least squares problem, the subspace predictive coefficients $L_w(k)$, $L_u(k)$, and $L_v(k)$ are derived as follows:

$$\min \left\| \mathcal{Y}_i^f(k) - [L_{w,i}(k)\ L_{u,i}(k)\ L_{v,i}(k)] \begin{bmatrix} \mathcal{Q}_i^p(k) \\ \mathcal{U}_i^f(k) \\ \mathcal{W}_i^f(k) \end{bmatrix} \right\|_{\mathrm{II}}^2 \tag{15}$$

where $\mathcal{Q}_i^p(k) = [\mathcal{Y}_i^p(k), \mathcal{U}_i^p(k), \mathcal{W}_i^p(k)]^T$ denotes the subspace prediction matrix corresponding to the past I/O trajectory.

According to (15), an optimal cost function is designed as follows:

$$\begin{aligned} J_i^*(k) &= \min \sum_{\tau=1}^{N} \| \hat{\mathcal{Y}}_i(k+\tau|k) \|_{Q_i}^2 + \sum_{\tau=1}^{N} \| \hat{\mathcal{U}}_i(k+\tau|k) \|_{R_i}^2 \\ &= \min \sum_{\tau=1}^{N} \Big(\| \triangle h_i(k+\tau|k) \|^2 + (\| \triangle v_i(k+\tau|k) \|^2 \\ &\quad + \| a_i(k+\tau|k) \|^2 \Big)_{Q_i} + \sum_{\tau=0}^{N-1} \| u_i(k+\tau|k) \|_{R_i}^2 \end{aligned} \tag{16}$$

where Q and R are positive definite weight matrices. Here, $\hat{\mathcal{U}}_{i,\min} \le \hat{\mathcal{U}}_i(k) \le \hat{\mathcal{U}}_{i,\max}$, $\hat{\mathcal{Y}}_{i,\min} \le \hat{\mathcal{Y}}_i(k) \le \hat{\mathcal{Y}}_{i,\max}$, with $\hat{\mathcal{Y}}$ and $\hat{\mathcal{U}}$ denote the optimal output and input sequences. The subscripts of max and min stand for the maximum and minimum allowable values.

Then, the feedforward control law of CAV becomes

$$\begin{aligned} \hat{\mathcal{U}}_i(k_\tau) &= \Big(\Gamma_i^T L_{u,i}^T(k) Q_i \Gamma L_{u,i}(k) + R_i \Big)^{-1} \Big(\Gamma_i L_{u,i}(k) \Big)^T \\ &\quad F_i \Big(\hat{\mathcal{Y}}_i(k) - L_{w,i}(k)\mathcal{W}_i^p(k) - L_{v,i}(k)\mathcal{W}_i^f(k) \Big) \\ &= K_i \mathcal{X}(k) \end{aligned} \tag{17}$$

where K_i is the control gain.

Eventually, the first element of control law (17) is implemented to steer the vehicular platoon.

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D. Platoon String Stability

To derive string stability conditions for the platoon, the platoon is expressed in a lumped form [40]. Based on the control law (17), system (4) is rewritten into a closed-loop form

$$\mathcal{X}_i(k+1) = \mathcal{A}_i\mathcal{X}_i(k) + \mathcal{E}_i\mathcal{W}_1(k) \tag{18}$$

with the system matrices

$$\mathcal{A}_i = \begin{bmatrix} 1 & -T_s & -\tau_i & 0 \\ 0 & 1 & T_s & 0 \\ 0 & 0 & (\tau_i - T_s)/\tau & 0 \\ (T_sK_{1,i})/\tau_i & (T_sK_{2,i})/\tau_i & (T_sK_{3,i})/\tau_i & -(T_s)/\tau_i \end{bmatrix}$$

$$\mathcal{E}_i = \begin{bmatrix} 0 & T_s & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \ \mathcal{X}_i(k) = \begin{bmatrix} \triangle h_i(k) \\ \triangle v_i(k) \\ \triangle a_i(k) \\ u_i(k) \end{bmatrix}$$

.

Based on (18), a lumped form of platoon model is deduced as follows:

$$\begin{cases} \bar{\mathcal{X}}(k+1) = \bar{\mathcal{A}}\bar{\mathcal{X}}(k) + \bar{\mathcal{E}}\mathcal{W}_1(k) \\ \bar{\mathcal{Y}}_i(k) = \bar{\mathcal{C}}_i\bar{\mathcal{X}}(k) \end{cases} \tag{19}$$

where $\bar{\mathcal{X}}(k) = [\mathcal{X}_2^T(k), \ldots, \mathcal{X}_M^T(k)]^T$, and matrices $\bar{\mathcal{A}}$ and $\bar{\mathcal{E}}$ have simliar definition to $\bar{\mathcal{X}}(k)$. $\bar{\mathcal{C}}_i = [0_{l\times n(i-1)}, \mathcal{C}_i, 0_{l\times n(M-i)}]$.

According to the Laplace transformation [41], (19) is transformed into a Laplace domain. That is

$$\widehat{\mathcal{Y}}_i(s) = \bar{\mathcal{C}}_i(sI - \bar{A})^{-1}e^{-\tau s}\bar{\mathcal{E}}\mathcal{W}_1(k) = \mathcal{P}_i(s)\mathcal{W}_1(k). \tag{20}$$

Then, the closed-loop control system (19) achieves ℓ_2 string stability if and only if the transfer function satisfies the following property:

$$\|\Gamma_i(z)\|_{H_\infty} = \sup_{\widehat{\mathcal{Y}}_{i-1}\neq 0} \frac{\|\Gamma_i(z)\widehat{\mathcal{Y}}_{i-1}(z)\|_{\ell_2}}{\|\widehat{\mathcal{Y}}_{i-1}(z)\|_{\ell_2}} \leq 1 \tag{21}$$

$$\|\widehat{\mathcal{Y}}_i(k)\|_{\ell_2}^2 = \sum_{n=-\infty}^{\infty} |\widehat{\mathcal{Y}}_i(k)|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |Y_i(e^{j\omega})|^2 d\omega \tag{22}$$

where Y_i stands for the discrete-time Fourier transform of $\widehat{\mathcal{Y}}_i(k)$. ω is the angular frequency of each sample. The output $\widehat{\mathcal{Y}}_i(k)$ is bounded as follows:

$$\begin{aligned} \|\widehat{\mathcal{Y}}_i(k)\|_{\ell_2}^2 &= \frac{1}{2\pi}\int_{-\pi}^{\pi} |Y_i(e^{j\omega})|^2 d\omega \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \left(\|\Gamma_i(e^{j\omega})\|_{H_\infty}|Y_{i-1}(e^{j\omega})|\right)^2 d\omega \\ &= \|\Gamma_i(e^{j\omega})\|_{H_\infty}^2 \|\widehat{\mathcal{Y}}_{i-1}(k)\|_{\ell_2}^2. \end{aligned} \tag{23}$$

Given that $\|\Gamma_i(e^{j\omega})\|_{H_\infty}^2$ is the tightest upper bound of $\|\mathcal{Y}_i(k)\|_{\ell_2}/\|\widehat{\mathcal{Y}}_{i-1}(k)\|_{\ell_2}$. Let us define $\|\widehat{\mathcal{Y}}_{i-1}(k)\|_{\ell_2}$ such that

$$|Y_{i-1}(e^{j\omega})| = \begin{cases} \sqrt{\dfrac{\pi}{2\epsilon}}, & \text{if } |\omega - \omega_0| < \epsilon \text{ or } |\omega + \omega_0| < \epsilon \\ 0, & \text{otherwise.} \end{cases} \tag{24}$$

Choosing ω_0 satisfies $\|\Gamma_i(e^{j\omega_0})\| = \|\Gamma_i(e^{j\omega})\|_{H_\infty}$, where Γ_i attains its maximum gain at this frequency, with ϵ approaching zero ($\epsilon \to 0^+$). To satisfy (23), $\widehat{\mathcal{Y}}_{i-1}(k)$ must adhere to Parseval's theorem, specifically ensuring that $\|\widehat{\mathcal{Y}}_{i-1}(k)\|_{\ell_2}^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |Y_{i-1}(e^{j\omega})|^2 d\omega = 1$. Thus

$$\begin{aligned} \|\widehat{\mathcal{Y}}_i(k)\|_{\ell_2}^2 &= \frac{1}{2\pi}\int_{-\pi}^{\pi} |Y_i(e^{j\omega})|^2 d\omega \\ &= \lim_{\epsilon\to 0^+} \frac{1}{2\pi}\left(\int_{-\omega_0-\epsilon}^{-\omega_0+\epsilon} |\Gamma_i(e^{j\omega})Y_{i-1}(e^{j\omega})|^2 d\omega \right. \\ &\quad \left. + \int_{\omega_0-\epsilon}^{\omega_0+\epsilon} |\Gamma_i(e^{j\omega})Y_{i-1}(e^{j\omega})|^2 d\omega\right) \\ &= \|\Gamma_i(e^{j\omega})\|_{H_\infty}^2 \frac{1}{2\pi}\int_{-\pi}^{\pi} |Y_{i-1}(e^{j\omega})|^2 d\omega \\ &= \|\Gamma_i(e^{j\omega})\|_{H_\infty}^2. \end{aligned} \tag{25}$$

Here, $\|\Gamma_i(e^{j\omega})\|_{H_\infty}^2$ serves as a rigorous upper bound for the proposed signal. Then, we can get that the minimal achievable upper bound for $\|\widehat{\mathcal{Y}}_i(k)\|_{\ell_2}/\|\widehat{\mathcal{Y}}_{i-1}(k)\|_{\ell_2}$.

Note that the leader-predecessor following (LPF) communication topology is implemented to the platoon. Actually, the reselection of vehicles from the source domain would not cause an obvious change in the transmission mode of communication links within the communication topology. It means such this phenomenon does not disrupt the stability of the platoon, and similar results can be found in the referenced literature [42].

IV. Experiments Design and Results

In this section, a hardware experimental platform is designed. Then, several comparative experiments are carried out to verify the advantage of the proposed DSTDC strategy.

A. Experimental Platform

The experiment is carried out on a vehicular platoon platform (see Fig. 3) comprising three vehicles, a WiFi router, seven ultra wide band (UWB) positioning modules. Such a platoon configuration with three vehicles has been employed in a similar study [43]. The key benefit of utilizing small mobile robots, in addition to CAVs, is their ability to simulate and test fundamental principles of autonomous movement and coordination in controlled environments. This approach enables us to explore the dynamics of multiagent systems, traffic flow, and vehicle-to-vehicle communication in a simplified context before applying these concepts to full-scale vehicular systems [44]. As depicted in Fig. 4, each vehicle is equipped with ROS system and nano processing unit. The proposed DSTDC strategy is deployed with C++ language on nano processing unit, enabling the generation of an optimal control input a_{des}. Since using a_{des} to drive the platoon is not feasible, a PID-based lower-level controller is employed to convert a_{des} into pulse width modulation (PWM) signals for platoon control. Furthermore, several key experimental parameters have been listed in Table I.

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TABLE I
EXPERIMENTAL PARAMETERS

Symbol	Description	Value	Symbol	Description	Value
N	Prediction horizon	15	L	Excitive order	10
T_s	Sampling period	10 Hz	h^*	Desired spacing	0.5 m
m_1, m_2, m_3	Vehicle mass	10 713 kg	τ	Inertial delay	0.1 s
$\triangle s_{\max}$	Maximum spacing	0.3 m	$\triangle s_{\min}$	Minimum spacing	−0.3 m
$\triangle v_{\max}$	Maximum velocity error	0.2 m/s	$\triangle v_{\min}$	Minimum velocity error	−0.2 m/s
$\triangle a_{\max}$	Maximum acceleration	0.03 m/s^2	$\triangle a_{\min}$	Minimum acceleration	−0.03 m/s^2

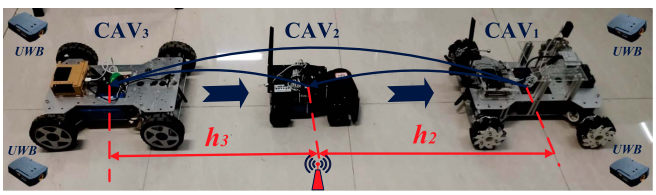

Fig. 3. Vehicular platoon hardware platform.

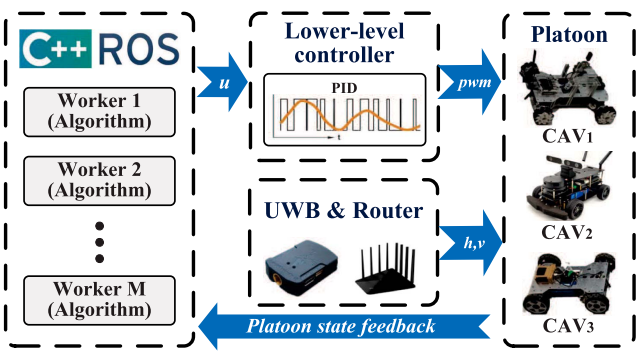

Fig. 4. Experimental framework.

B. Experimental Procedures

The platoon configuration with three vehicles traveling on an off-ramp road. Here, the desired safe spacing of platoon is set to 0.5 m. The source vehicle CAV_1 leaves the platoon at 18 s, and another source vehicle would be selected to guide the platoon.

The experiment process consists of two stages: data collection and strategy verification. During the data collection stage, the results of the traditional model predictive control strategy [7] are employed to train the data-driven model of source domain vehicle. Additionally, three sets of experiments are conducted to validate the effectiveness of the proposed strategy, detailed as follows:

1) *Exp1*: The data-driven predictive control (DPC) strategy [32] is implemented to platoon. In this case, the platoon subjects to the communication topology 1.
2) *Exp2*: The transferable data-driven control (TDC) strategy is implemented to platoon under communication topology 2. Here, the platoon would not change the source domain vehicle while CAV_1 leaves the platoon.
3) *Exp3*: The proposed DSTDC strategy is implemented to platoon control. Under communication topology 2, the platoon would reselect the source domain vehicle while CAV_1 leaves the platoon.

In this experiment, we collected experimental data $\mathbb{D} \in \mathbb{R}^{4\times 3}$ from 4 I/O trajectories $[\triangle h_i(k), \triangle v_i(k), a_i(k), u_i(k)]$ of three vehicles. Note that this dataset is used to construct the data-driven model. Furthermore, different control strategies are embedded into the platoon, and similar data sets were collected to validate the effectiveness of the proposed algorithm.

V. RESULTS AND DISCUSSIONS

The platoon tracking performance and computational efficiency of different strategies are analyzed in this section.

A. Platoon Tracking Performance

The platoon tracking performance is evaluated in terms of the spacing, velocity and acceleration. Figs. 5 and 6 present the results of spacing and velocity responses of CAV_2 and CAV_3, which subject to different control strategies. TDC and DSTDC strategies exhibit obvious fluctuations of spacing and velocity, particularly within 0–4.5 s. This phenomenon is expected due to CAV_2 and CAV_3 needing to initialize the data-driven models after receiving knowledge from source domain vehicles. Here, TDC strategy executes an oscillation at 18 s, it causes by the source domain vehicle (CAV_1) leaving the platoon. In contrast, DSTDC strategy selects a new source domain vehicle, and quickly stabilizes the spacing and velocity to a consistent range. Furthermore, DPC strategy outperforms both TDC and DSTDC in terms of spacing and velocity tracking. This is because each vehicle with DPC strategy relies on its own learned knowledge rather than the source domain vehicle. Therefore, we can conclude that the proposed DSTDC strategy could maintain the performance of platoon under off-ramp road.

As depicted in Fig. 7, the results of acceleration with different strategies are analyzed. The DPC strategy shows a relatively narrow range of acceleration for platoon, with most data points concentrated around the median. TDC, represented in black, exhibits a wider range of acceleration values, suggesting greater variability in response to traffic conditions. This trend might lead to unstable driving behavior, potentially causing uncomfort. Note that DSTDC displays a performance that lies between DPC and TDC. Although it exhibits a wider acceleration range than DPC, it remains more limited compared to TDC, suggesting a balanced strategy in managing acceleration variations.

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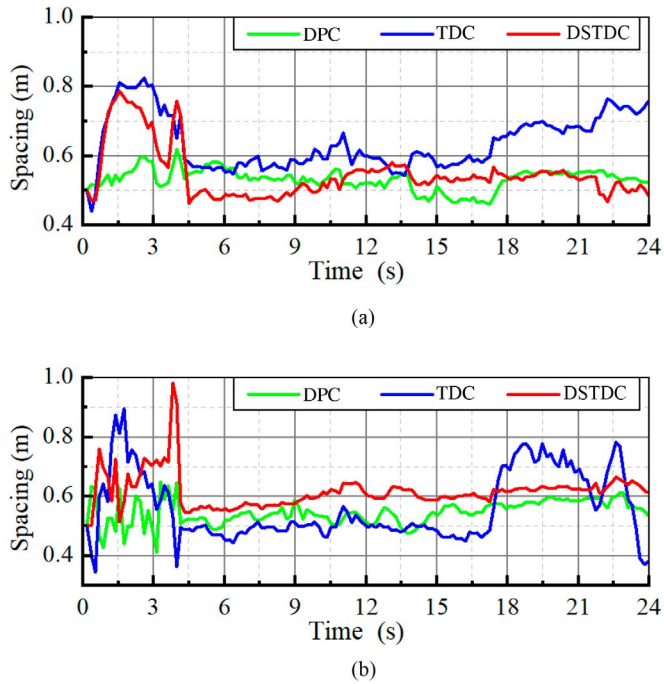

Fig. 5. Results of the spacing with different strategies. (a) Spacing of CAV_2. (b) Spacing of CAV_3.

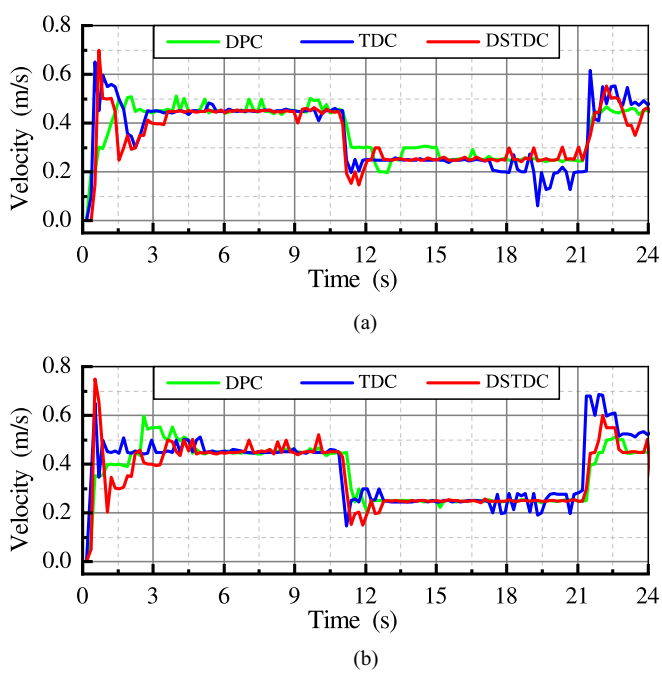

Fig. 6. Results of the velocity with different strategies. (a) Velocity of CAV_2. (b) Velocity of CAV_3.

To elaborate on the advantages of our proposed DSTDC strategy, the absolute maximum error (AMAXE) and RMSE assessment indexes are introduced. With this in mind, the spacing (h), velocity error ($\triangle v$) and acceleration (a) of platoon are presented in Table II. The DPC strategy has the minimum AMAXE and RMSE, which implies the desirable platoon stability could

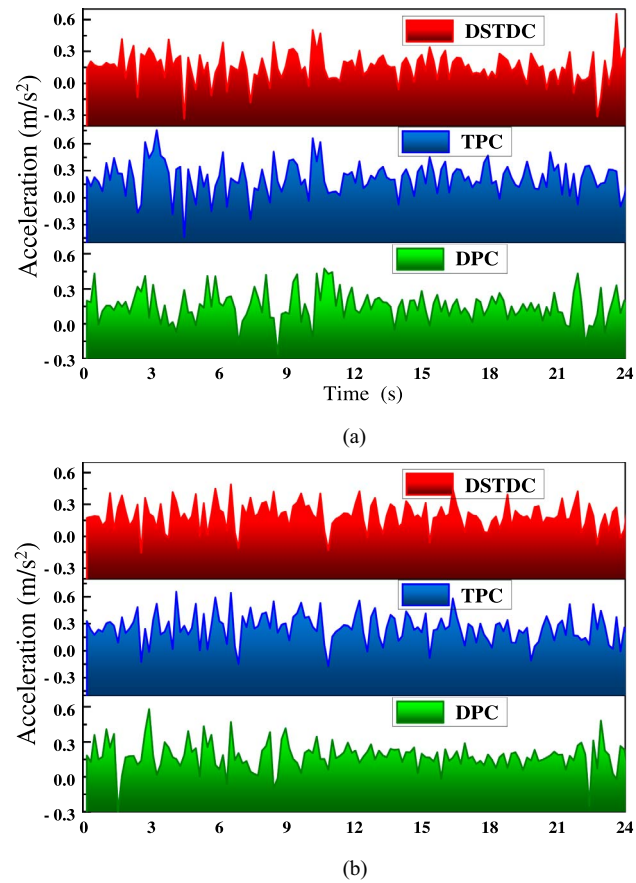

Fig. 7. Results of the acceleration with different strategies. (a) Acceleration of CAV_2. (b) Acceleration of CAV_3.

be ensured. Furthermore, TDC strategy executes the maximum AMAXE and RMSE, and DSTDC strategy falls between the aforementioned methods.

B. Computational Efficiency

The computational efficiency of the algorithm is crucial for platoon. Here, the convergence time and computational time of each solving process are used to evaluate the performance of different strategies. Fig. 8 illustrates the TDC and DSTDC strategies execute superior convergence performance within four iteration steps than DPC. This phenomenon is attributed to the knowledge of CAV_1 is transferred to CAV_2 and CAV_3, thereby eliminating the initialization process.

The average computational time of each sampling interval among three strategies is presented in Fig. 9. Specifically, TDC and DSTDC require around 0.027 s and 0.028 s for CAV_2, and approximately 0.026 s and 0.028 s for CAV_3. In contrast, DPC requires significantly more time, with 0.037 s for CAV_2 and 0.039 s for CAV_3. This indicates DSTDC strategy executes a 27.2% improvement in solving efficiency compared to DPC strategy. These results suggest that both TDC and DSTDC are more efficient in terms of computational resource consumption, which is crucial for real-time applications in platoon.

Furthermore, the transfer rate and parameter divergence [45] have also been analyzed to investigate the effectiveness of

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TABLE II
EXPERIMENTAL RESULTS ANALYSIS WITH AMAXE AND RMSE

Symbol		AMAXE			RMSE		
		h (m)	$\triangle v$ (m/s)	a (m/s)^2	h (m)	$\triangle v$ (ms)	a (m/s)^2
DPC	CAV_2	0.618	0.013	0.471	0.030	0.004	0.131
	CAV_3	0.650	0.097	0.581	0.041	0.014	0.122
TPC	CAV_2	0.825	0.051	0.754	0.079	0.032	0.183
	CAV_3	0.896	0.085	0.656	0.072	0.031	0.162
DSTDC	CAV_2	0.787	0.099	0.553	0.054	0.012	0.140
	CAV_3	0.782	0.049	0.489	0.059	0.022	0.125

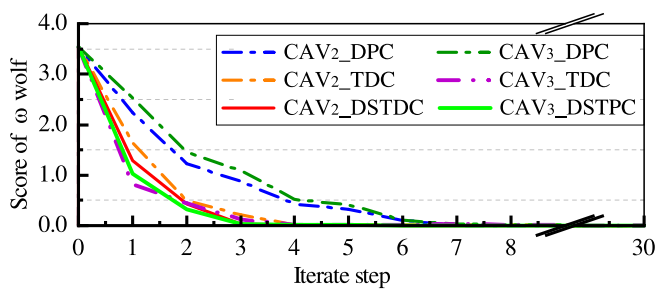

Fig. 8. Results of convergence performance.

TABLE III
EXPERIMENTAL RESULTS OF TRANSFER RATE AND PARAMETER DIVERGENCE

Symbol	TDC		DSTDC	
	CAV_2	CAV_3	CAV_2	CAV_3
Tranfer rate	68%	61%	48%	46%
Parameter divergence	0.18	0.21	0.23	0.24

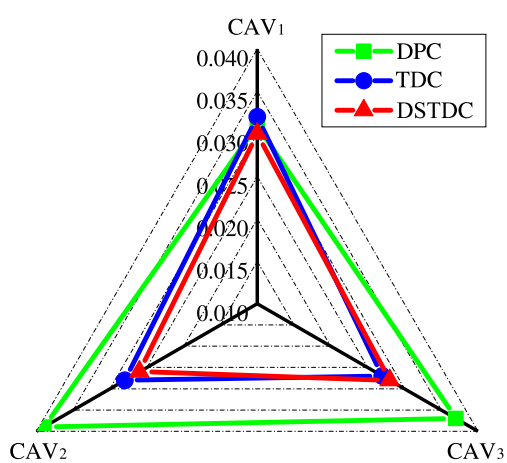

Fig. 9. Results of average computational time.

transfer learning in Table III. Here, transfer rate refers to the effectiveness of transferring knowledge from the source domain to the target domain, and parameter divergence represents the difference in model parameters between the source and target domains. Note that a higher transfer rate indicates a higher probability of success, and a lower parameter divergence is better for the platoon. CAV_2 with TDC achieves a transfer rate of 68% and a parameter divergence of 0.18. Likewise, TDC with CAV_3 shows a transfer rate of 61% and a parameter divergence of 0.21, outperforming the results of DSTDC. It implies that TDC has more effective in transferring data with minimal discrepancies, ensuring a better transfer performance. In this experiment, the source vehicle CAV_1 leaves the platoon at 18 s, while the dynamic source-aware mechanism is activated. Then, we CAV_2 is assigned as the new source vehicle in terms of the data accumulation, computing performance and driving performance indicators. Additionally, the transfer rate is positively correlated with these metrics, which explains why CAV_2 exhibits the highest transfer rate compared to CAV_3. However, it is essential to balance transferring performance with safety considerations, as DSTDC offers enhanced platoon safety.

Therefore, it can be concluded that the TDC strategy could hold computational efficiency but struggles to handle complex scenarios where the source domain vehicle departs the platoon. In contrast, the DSTDC strategy improves computational efficiency while addressing platoon tracking issues in intricate road conditions. Furthermore, we also observe several notable findings. According to the experimental results, we can get that the transfer learning method is effective in improving computational efficiency only for the target domain vehicles, with minimal impact on the source domain vehicle. Additionally, there is a positive correlation between the accuracy of source domain selection and computational cost, indicating that reducing redundant data can substantially speed up the training process.

VI. CONCLUSION

In this article, we have developed an DSTDC to mitigate the computational demands associated with platoon control in off-ramp situations. The subspace identification method has been utilized to establish the data-driven model in terms of I/O trajectories, it could alleviate the adverse effects caused by the complex powertrain structure of vehicular platoon. Then, a transfer learning approach has been applied to the source

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domain vehicle of platoon to reduce the computational burden. With this in mind, the target domain vehicle should just update the knowledge in terms of the individual dynamic characteristics and tasks. Furthermore, an adaptive switch mechanism has been developed to address the issue of source domain vehicle departure in the off-ramps. Finally, the experimental results indicate that our proposed DSTDC strategy could provide satisfactory platoon tracking performance and improve 27.2% computational efficiency. Furthermore, limited experimental scenarios and datasets make it challenging to validate the generalizability of the proposed strategy on complex scenarios. In future work, we will extend the scale of the vehicle platoon with the advanced vehicle dynamics models, incorporating a larger number of actual vehicles to construct more realistic traffic conditions. In addition, stability theory analysis is also a direction we need to strengthen.

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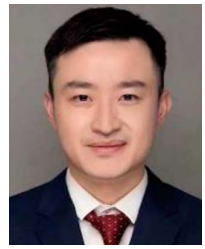

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